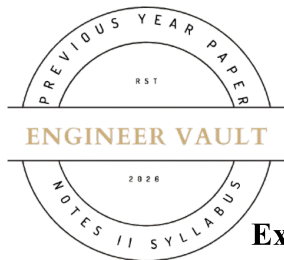


Total No. of Printed Pages: 2

**SUBJECT CODE NO: - RNP-8024****FACULTY OF SCIENCE AND TECHNOLOGY****S.E (CSE/CS/CE/AI&DS/AIML/IT/AI) (NEP) Sem - III****Examination November / December 2025 (To be held in February-2026)****Discrete Mathematics****[Time:2:00 Hours]****[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B

- 1) Part A: Question No 1 is compulsory.
- 2) Attempt any four questions from remaining Part B: Q.2 to Q.7.

PART A

Q1

Solve any 5 following questions

10

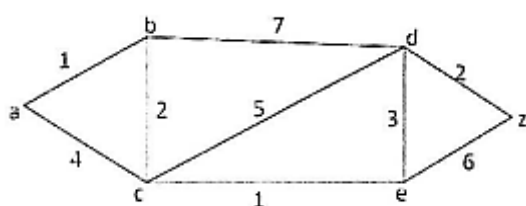
- a) The number of elements in $p\{p(A)\}$ of a set $A = \{1, \{2,3\}\}$ is
a) 16 b) 4 c) 8 d) None of these
- b) Let $f(x)=x^2+x$ and $g(x)=x+1$ then fog
a) x^2+x+1 b) x^2-x+1 c) x^2+3x+2 d) None of these
- c) In how many ways can be committee 2 men and 4women be selected from 7 men and 9 women?
a) ${}^7C_2 \times {}^9C_4$ b) 2646 c) 4626 d) both a and b
- d) A tree in which one vertex is distinguished from all the other vertices is called _____
a) Rooted tree b) Simple tree c) M-ary tree d) None of these.
- e) Which of the following is correct?
a. Every integral domain is ring
b. Every field ring
c. Every field is an integral domain
d. all of these.
- f) A simple connected graph without any circuits is called _____
a) Forest b) bipartite graph c) tree d) None of these.
- g) The general solution of recurrence relation is _____
a. The Difference of homogeneous and particular solutions.

- b. The Sum of homogeneous and particular solutions.
- c. The product of homogeneous and particular solutions
- d. None of these.

PART B

Solve any four from the following

- Q2 Prove the De-morgan's laws of set theory. 05
- Q3 Explain different types of functions with suitable examples. 05
- Q4 Describe different operations on graphs with example. 05
- Q5 Find the Total solution of $a_r - 8a_{r-1} + 16a_{r-2} = 0$ with $a_2 = 32, a_3 = 64$ 05
- Q6 Apply Dijkstra's shortest path algorithm to find the shortest path between the vertices a and z in the figure below. 05



- Q7 Let $(\{a,b\}, *)$ be a semi group where $a*a=b$ show that: 05
- i) $a*b=b*a$
 - ii) $b*b=b$

